

A c_0 -VECTOR OUTSIDE THE CLOSED SPAN IN A VARIABLE-EXPONENT SEQUENTIAL SPACE

AGENT LANE 05

SOURCE AND TARGET

The source paper is Talponen, *Constructions of sequential spaces*, arXiv:0905.0812 [1]. It introduces the sequential variable-exponent spaces

$$\ell^{p(\cdot)} = \{x \in \ell^\infty : \Phi(x) < \infty\},$$

where, for $p : \mathbb{N} \rightarrow [1, \infty]$,

$$|||x|||(1) = |x_1| \boxplus_{p(1)} |x_2|, \quad |||x|||(k) = |||x|||(k-1) \boxplus_{p(k)} |x_{k+1}|,$$

and $\Phi(x) = \lim_k |||x|||(k)$. Here $a \boxplus_r b = (a^r + b^r)^{1/r}$ for finite r .

On page 8 the paper repeats the following open problem:

Does $\ell^{p(\cdot)} \cap c_0$ always coincide with $[(e_n)] \subset \ell^{p(\cdot)}$?

$$\lim_{n \rightarrow \infty} d_{\text{BM}}(Q_n(\ell^{p(\cdot)}), \ell^1) = 1.$$

Finally we reiterate the open problems that have arisen in this note:

- We do not know if the space in Example 3.1 is MLUR.
- Does $\ell^{p(\cdot)} \cap c_0$ always coincide with $[(e_n)] \subset \ell^{p(\cdot)}$?
- Given $p : \mathbb{N} \rightarrow (0, \infty)$, is $\ell^{p(\cdot)}$ necessarily strictly convex or smooth?

RESULT

The answer is negative. We construct a finite-valued exponent map $p : \mathbb{N} \rightarrow [2, \infty)$ and a vector $x \in \ell^{p(\cdot)} \cap c_0$ such that $x \notin [(e_n)]$.

PROOF INTUITION

The obstruction is that membership in the closed span $[(e_n)]$ requires the coordinate tails $Q_n x = x - P_n x$ to vanish in the $\ell^{p(\cdot)}$ norm. We build x from blocks. Inside block j the exponent is very large, $r_j = j + 1$, and there are j^{r_j} equal coordinates c/j . Viewed by itself, the block has norm exactly c , so tails starting at the block never become small. Viewed after the previously accumulated initial segment, however, the same block contributes only c^{r_j} under an r_j -root. Since $\sum_j c^{r_j}/r_j < \infty$, the global recursive norm remains bounded.

Lemma 1. *For every $p : \mathbb{N} \rightarrow [1, \infty]$ and every $y \in \ell^{p(\cdot)}$,*

$$y \in [(e_n)] \iff \|Q_n y\|_{\ell^{p(\cdot)}} \rightarrow 0.$$

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Proof. The implication from right to left is immediate because $P_n y$ is finitely supported and

$$y - P_n y = Q_n y \rightarrow 0.$$

Conversely, suppose $y \in [(e_n)]$. Given $\varepsilon > 0$, choose a finitely supported vector z , supported in $\{1, \dots, N\}$, such that $\|y - z\|_{\ell^{p(\cdot)}} < \varepsilon$. For $n \geq N$, $Q_n z = 0$. The recursive norm is coordinatewise monotone in the absolute values of the coordinates, so every coordinate projection, including Q_n , is contractive. Hence

$$\|Q_n y\|_{\ell^{p(\cdot)}} = \|Q_n(y - z)\|_{\ell^{p(\cdot)}} \leq \|y - z\|_{\ell^{p(\cdot)}} < \varepsilon$$

for all $n \geq N$. □

Theorem 2. *There are $p : \mathbb{N} \rightarrow [2, \infty)$ and*

$$x \in \ell^{p(\cdot)} \cap c_0$$

such that $x \notin [(e_n)]$. Therefore $\ell^{p(\cdot)} \cap c_0$ need not coincide with $[(e_n)] \subset \ell^{p(\cdot)}$.

Proof. Fix $0 < c < 1$, for instance $c = 1/2$. Put

$$r_j = j + 1, \quad m_j = j^{r_j} \quad (j \geq 1).$$

Let $N_1 = 2$ and $N_{j+1} = N_j + m_j$. The j -th block of coordinates is

$$B_j = \{N_j, N_j + 1, \dots, N_{j+1} - 1\}.$$

Define $x \in \ell^\infty$ by

$$x_1 = 1, \quad x_n = \frac{c}{j} \quad \text{if } n \in B_j.$$

Since $c/j \rightarrow 0$, $x \in c_0$.

Define $p : \mathbb{N} \rightarrow [2, \infty)$ by

$$p(k) = r_j \quad \text{whenever } N_j - 1 \leq k \leq N_{j+1} - 2.$$

These intervals partition $\mathbb{N}$, so p is well defined.

Let A_j be the norm of the initial segment consisting of x_1 and the first j whole blocks; equivalently

$$A_j = \|P_{N_{j+1}-1} x\|_{\ell^{p(\cdot)}}.$$

Then $A_0 = 1$. Since every exponent used while appending block j is r_j ,

$$A_j = \left(A_{j-1}^{r_j} + m_j \left(\frac{c}{j} \right)^{r_j} \right)^{1/r_j} = \left(A_{j-1}^{r_j} + c^{r_j} \right)^{1/r_j}.$$

As $A_{j-1} \geq 1$,

$$A_j = A_{j-1} \left(1 + \left(\frac{c}{A_{j-1}} \right)^{r_j} \right)^{1/r_j} \leq A_{j-1} (1 + c^{r_j})^{1/r_j}.$$

Therefore

$$A_j \leq \prod_{i=1}^j (1 + c^{r_i})^{1/r_i}.$$

The infinite product converges because

$$\sum_{i=1}^{\infty} \frac{\log(1 + c^{r_i})}{r_i} \leq \sum_{i=1}^{\infty} \frac{c^{r_i}}{r_i} < \infty.$$

All finite prefixes inside a block have norm at most the norm after the whole block. Hence $\sup_n \|P_n x\|_{\ell^{p(\cdot)}} < \infty$, so $x \in \ell^{p(\cdot)}$.

It remains to show that $x \notin [(e_n)]$. Consider the tail starting just before block j , namely $Q_{N_j-1}x$. The finite projection of this tail onto the coordinates in B_j has all earlier coordinates equal to zero, then m_j coordinates equal to c/j , and all relevant exponents equal to r_j . Its norm is

$$\left(m_j \left(\frac{c}{j} \right)^{r_j} \right)^{1/r_j} = c.$$

By coordinatewise monotonicity,

$$\|Q_{N_j-1}x\|_{\ell^{p(\cdot)}} \geq c$$

for every j . Thus the tails of x do not tend to zero in norm. Lemma 1 gives $x \notin [(e_n)]$. $\square$

VERIFICATION NOTES

The construction uses only finite exponents $p(k) \geq 2$, so it fits the source paper’s setting $p : \mathbb{N} \rightarrow [1, \infty]$, and also the broader $(0, \infty)$ -valued wording of the final open-problem list.

A small script, `code/block_counterexample_check.py`, prints the first few values of the block recurrence

$$A_j = (A_{j-1}^{j+1} + c^{j+1})^{1/(j+1)}$$

and the fixed lower bound c for the selected block tails. This is only a sanity check for the displayed formulas; the convergence proof above is analytic.

NOVELTY AND SEARCH BOUNDS

Before promotion, the run’s registry, solution, attempt, and proof-gap indexes were searched for arXiv:0905.0812, the paper title, Talponen, and the core phrases around $\ell^{p(\cdot)} \cap c_0$ and $[(e_n)]$. No existing solution packet or serious prior attempt for this question was found.

External web searches on 2026-06-15 used exact and near-exact phrases including

“Constructions of sequential spaces” Talponen “ c_0 ” “ $[(e_n)]$ ”,

“ $\ell^p(\cdot)$ ” “ c_0 ” “ $[(e_n)]$ ”,

and

“variable exponent sequential spaces” Talponen.

These searches surfaced the source paper but no later explicit answer to this equality question. The packet should therefore be read as a candidate new

counterexample with medium novelty confidence, pending broader human literature review.

HUMAN REVIEW RECOMMENDATION

Review as a candidate full negative answer to the second open problem in [1]. The main checks are the indexing of the block exponents, the contractivity of coordinate projections under the recursive norm, and the tail-norm characterization in Lemma 1.

REFERENCES

- [1] J. Talponen, *Constructions of sequential spaces*, arXiv:0905.0812, 2009.

Human Review: Variable-Exponent c_0 Tail Counterexample

Original automatic proof: `solution_packet.pdf`

Checked date: 15 June 2026

Status: Manually checked.

Verdict: The proof appears correct.

Human Comments

The proof appears to give a correct negative answer to Talponen's question asking whether

$$\ell^{p(\cdot)} \cap c_0$$

always coincides with the closed linear span

$$[(e_n)] \subset \ell^{p(\cdot)}.$$

Here $[(e_n)]$ means the closed linear span of the canonical unit vectors with respect to the $\ell^{p(\cdot)}$ -norm, not the c_0 - or ℓ_∞ -norm.

The construction is quite nice. The vector is built in blocks. In block j , the exponent is large,

$$r_j = j + 1,$$

and the block contains

$$m_j = j^{r_j}$$

coordinates, each equal to c/j , where $0 < c < 1$. Since $c/j \rightarrow 0$, the resulting vector belongs to c_0 .

The key idea is that each block has two different behaviours depending on how it is viewed. If the block is viewed by itself, starting from zero, its norm is exactly

$$\left(m_j \left(\frac{c}{j} \right)^{r_j} \right)^{1/r_j} = c.$$

Thus tails beginning just before block j always contain a finite piece of norm c . Consequently the tails cannot converge to zero in the $\ell^{p(\cdot)}$ -norm, and this prevents the vector from lying in $[(e_n)]$.

On the other hand, when the same block is appended after the previous nonzero part of the vector, the recursive norm gives

$$A_j = \left(A_{j-1}^{r_j} + c^{r_j} \right)^{1/r_j}.$$

Since $r_j \rightarrow \infty$ and $0 < c < 1$, the extra contribution c^{r_j} becomes very small. The estimate

$$A_j \leq A_{j-1}(1 + c^{r_j})^{1/r_j}$$

leads to the convergent product

$$\prod_j (1 + c^{r_j})^{1/r_j},$$

because

$$\sum_j \frac{c^{r_j}}{r_j} < \infty.$$

This shows that the norms of the finite initial segments are uniformly bounded, hence the vector belongs to $\ell^{p(\cdot)}$.

The auxiliary tail characterization also appears correct:

$$y \in [(e_n)] \iff \|Q_n y\|_{\ell^{p(\cdot)}} \rightarrow 0.$$

The proof uses only approximation by finitely supported vectors and contractivity of coordinate projections. The contractivity follows from coordinatewise monotonicity of the recursive norm, since coordinate projections only decrease the absolute values of coordinates.

Review Outcome

Archive this packet as an apparently correct counterexample. The proof's main mechanism is sound: each block remains large when seen as a fresh tail, but contributes less and less when appended after the previously accumulated part of the vector. This gives

$$x \in \ell^{p(\cdot)} \cap c_0 \quad \text{but} \quad x \notin [(e_n)].$$

I would recommend checking with the author whether the question has since been answered elsewhere, but mathematically the proposed construction appears correct.